# Design Notebook

Wednesday 1/21

- Everyone present
- Finalized student pick up system project idea with Professor
- Discussed our options for the system including facial recognition, license plate, QR scanner, and pin input.
- Identified potential issues like wireless connections, database design, and potential layout for the system. We will need to go to the school and view their current system to start designing the final layout.

Wednesday 1/28

- Haya was at the career fair, but everyone else was here.
- Wrote questions to ask people involved in the pickup process (teachers and parents).
- Wrote needs and objective statements to be finalized next Wednesday.
- The professor will plan a meeting with elementary school staff next week to discuss issues with the current system and requirements for implementation.

Monday 2/2

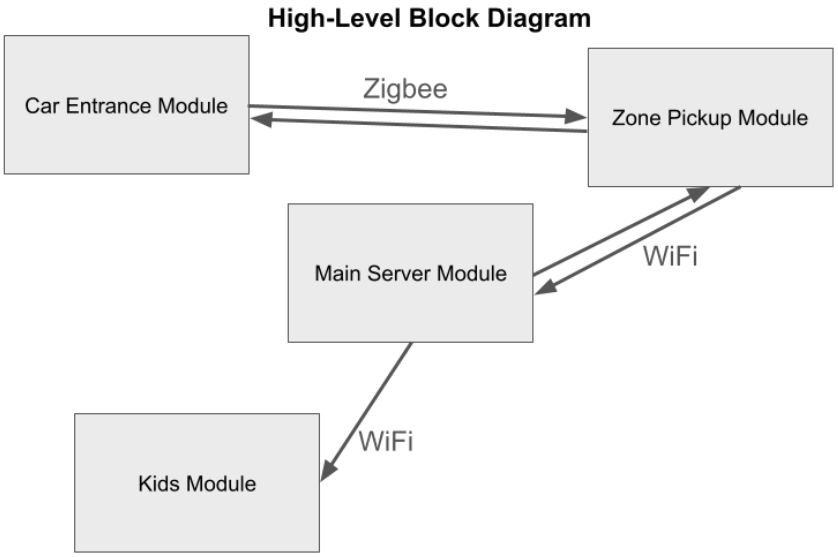
- Everyone present
- Near finalized needs and objective statements
- A meeting with school will need to happen before statements, but we are not sure when that will happen.
- Started work on our requirements for the project. We have marketing and engineering requirements, but need justification for each requirement.
- Our group did more research on hardware components, but without a meeting, it is hard to finalize anything.

Wednesday 2/4

- Everyone present
- Needs and objective statements were looked at by the professor and TA, revised based on the feedback they gave, then finalized and turned in.
- We wrote the justifications for all the engineering and marketing requirements based on the components and design choices we have made so far.
- The constraints and standards based on the environment that we are in were also written. We will likely get those and the requirements finalized next Monday.
- With the proposal being due next Friday, we split up the work that needs to be done between us so that we can work on it without needing to be in a meeting.

Monday 2/9

- Everyone present
- We had the professor go over our requirements for our project and adjusted the requirements based on his feedback.
- Most of the time today was spent working on our individual parts for the proposal document and presentation.
- We finished the meeting by doing some preparations for our presentation on Wednesday, assigning slides to each team member.
- Jack also went with the TA to look for hardware that we can use for our project

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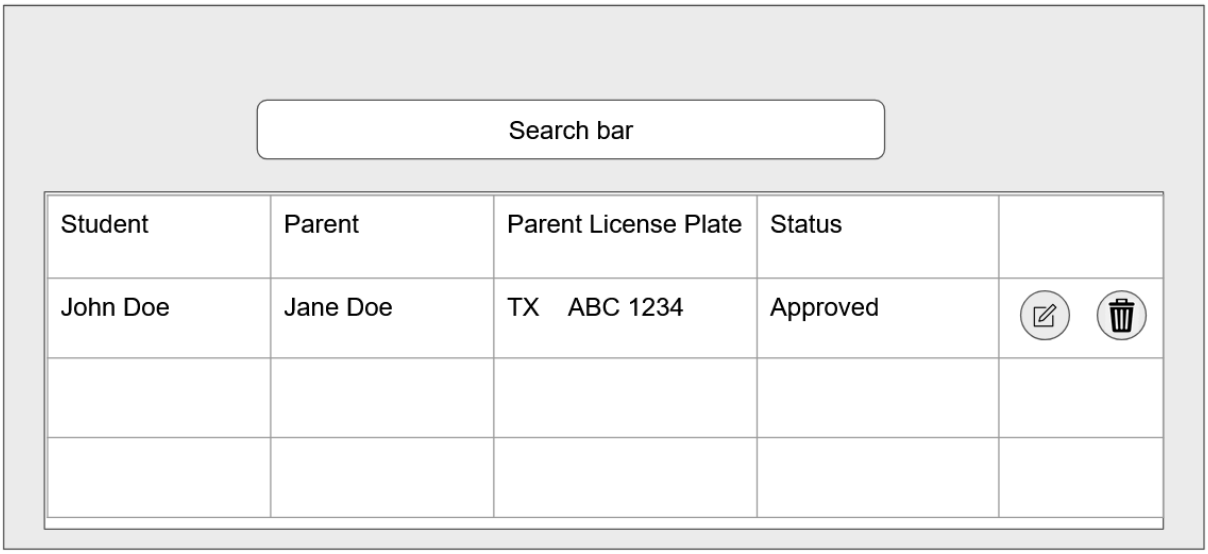
Wednesday 2/11

- Everyone present
- We did our proposal presentation today and received feedback on what to improve
- We watched the other groups present their projects
- We revised our presentation slides based on the feedback and resubmitted our slides

Monday 2/16

- Everyone present
- We setup our methods of communication (Github, Discord, Drive)
- We went over our proposal presentation with the professor and modified our design based on his feedback.
- We started ordering parts after getting approval from the professor. Currently, we have only ordered parts that we know we will need soon.
- We began designing the website we will use for our project, going over what information we need from parents and students, modifying our database as well.

## Admin Screen- home



Wednesday 2/18

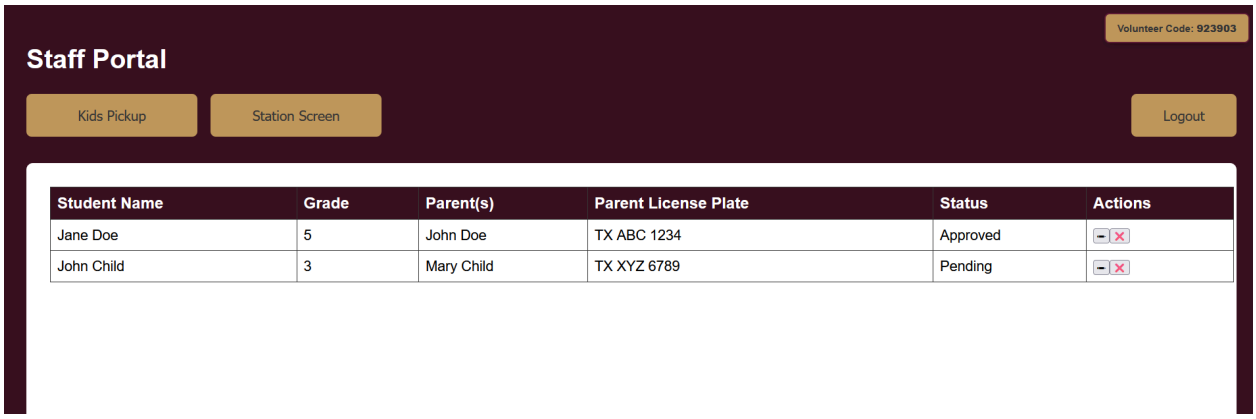
- Everyone present
- Website wireframes and initial layouts were finalized. We also selected the colors that will be used for our website.
- We started testing the camera that will be used to recognize either the license plate or a QR code, we are discussing both options
- Setup a new way to track project progress as our free trial ran out on the site we were using.
- We also set up a project on a website that will allow us to design the mount for the camera, display and the Raspberry Pi.

Monday 2/23

- Everyone present
- We had a short meeting with the professor, however we did not have much to go over with him yet as not much of the project has been completed or difficult either.
- We worked on the actual coding part for the website frontend using placeholder data and tables as the database is not yet connected to the database.
- Jack got the camera working again and we will begin testing the camera on Wednesday.

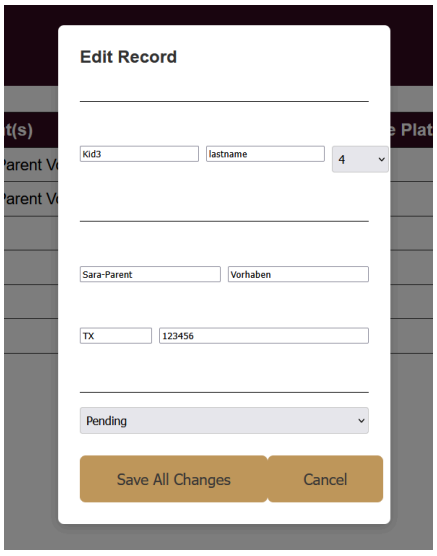
Wednesday 2/25

- Everyone present
- Further worked on the frontend of the website.
- Started getting Zigbee setup.
- Discussed our further project options




Monday 3/2

- Everyone present
- Haya, Sara and I worked on the backend for our website.
- Jack started testing image recognition software for the license/QR recognition.
- The meeting with the professor was very short as we have a lot of things to build and work on and we have not run into any roadblocks.



Wednesday 3/4

- Everyone present
- Really just a workday. Nothing new needed to be added to the plan during the meeting on Monday
- Haya, Sara and I continued to work on the backend related features for the website.
- Jack continued testing methods to better recognize license plates and Jayden continued to work on the physical and hardware parts of the project.

Monday 3/16

- Everyone present
- I worked on the CDR report
- Sara started working on QR code APIs and how to implement that.
- Met with the professor about our presentation.

Wednesday 3/18

- Remote workday
- Worked on an efficiency algorithm to optimize the car rider line system.

Monday 3/23

- Everyone present
- We met for our weekly meeting with the professor.
- He suggested that we meet with a teacher from an elementary school who is involved in the pickup process to get further insight on how the process works.
- We started working on getting the server for our project setup while Jack continued working on the license plate recognition

Wednesday 3/25

- Remote workday

Monday 3/30

- Everyone present
- We met with the professor and we went over changes to the website and discussed possible design changes for the housing that the Raspberry Pi and screen will sit in.

- I worked on getting the delete feature working and deletes the child from the database

Wednesday 4/1

- Everyone present
- We had a call with someone from the elementary school who is directly involved with the car rider system at the school who provided some more specific insight into the system
- I continued working on the delete function

Monday 4/6

- Everyone present
- Had our weekly meeting with professor, but nothing really important needed to be gone over
- I finished the delete function
- Jack is continuing to work on license plate recognition
- Jayden finished the housing re-design
- Sara and Haya have been working on the server and trying to get it communicating

Wednesday 4/8

- Everyone present
- Individual workday, but we still met to see if the server was working
- We were able to get the server working and connected to the website

Monday 4/13

- Everyone present
- We had our weekly meeting with the professor, but nothing really needed to be gone over as we are just working.
- I implemented the database queries using the data from the qr code

Wednesday 4/15

- Everyone present
- I wrote a script to constantly take and process pictures for our qr code reader
- We were able to get the server to communicate with the Pi and the data being sent can be processed by the server and the correct outputs are displayed.

Wednesday 4/22

- Everyone present
- We demoed our project for the professor and he liked what we have, but recommended some minor changes/improvements.